

Project Proposal

Title:

Following The Prompt: An empirical analysis of AI traffic routing across heterogeneous ISPs

Team members:

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1. Research Question

- RQ1: How does the choice of a local ISP (e.g., residential broadband versus mobile carrier) influence the Autonomous System (AS) path taken by traffic destined for major AI services (ChatGPT, Gemini, Claude)?
- RQ2: Assuming initial path divergence at the local ISP level, do AI queries ultimately converge on a centralized set of transit or CDN edge ASNs, potentially creating structural chokepoints and surveillance vulnerabilities?
- RQ3: Does the geographic routing of these AI queries cause data to unnecessarily cross international or distinct jurisdictional boundaries, and how does this vary based on the originating network?

2. Hypothesis

We hypothesize that while user queries to AI services appear direct, traffic originating from different local ISPs will traverse highly different paths due to differing commercial peering agreements, occasionally crossing distinct jurisdictional boundaries. Furthermore, we hypothesize that despite initial path divergence at the local ISP level, traffic from all studied networks will ultimately converge onto a highly centralized set of CDN edge ASNs (e.g., Cloudflare, Google Edge), creating single points of infrastructure dependency and potential data interception.

3. Contribution

This project provides empirical visibility into the hidden infrastructure dependencies of everyday AI tool usage. By mapping the physical and logical paths of AI queries across various ISPs, this research contributes concrete data to discussions around data sovereignty, highlighting how local network routing policies directly impact a user's global privacy footprint and vulnerability to network-level surveillance.

4. Methodology

We will conduct active network measurements over a fixed period of time using a Python-based automation script.

Data Collection

From distinct local networks (like college Wi-Fi, residential Xfinity, T-Mobile broadband), the script will execute `traceroute` and DNS lookups to the API endpoints and web interfaces of ChatGPT, Gemini, and Claude. However, relying solely on these three local networks available with us presents a methodological risk of geographic and network bias, as it may not yield a comprehensive dataset, limiting our ability to infer broader routing behaviors. To mitigate this risk and strengthen the validity of our research, we will supplement our local measurements by utilizing the RIPE Atlas network. By querying RIPE Atlas probes, we will execute parallel traceroutes from diverse global vantage points, significantly expanding our overall dataset beyond our three local networks to capture a truly comprehensive representative map of AI traffic routing.

Analysis

The collected trace data will be reviewed to map the general routing pathways from each ISP to the target AI services. We will use BGP (Border Gateway Protocol) data to map these individual IP hops into their larger Autonomous Systems(ASes). From there, we will conduct a broad review of the network hops to observe high-level trends in traffic distribution. We will also explore the data for potential variations in geographic routing, specifically monitoring for anomalous jurisdictional traversal and investigate whether there are common regional or infrastructural convergence points across the different connection types.

5. Biggest Risk

The primary risk to this research is the potential for geographic and network bias inherent in relying solely on the three local networks currently available University Wi-Fi, residential Xfinity, and T-Mobile broadband which may not yield a globally comprehensive dataset or allow for the inference of broader routing behaviors. To mitigate this, the study will incorporate the RIPE Atlas network to execute parallel traceroutes from diverse international vantage points, establishing a robust comparative baseline.

Furthermore, relying on historical BGP snapshots risks missing real-time network shifts. This is complicated by the fact that actual physical traffic paths frequently diverge from publicly advertised BGP routes. This assessment is based on our understanding of the methodology. We expect to refine our analytical approach as we explore more.